T5 — Energy Transfers in and Between Organisms

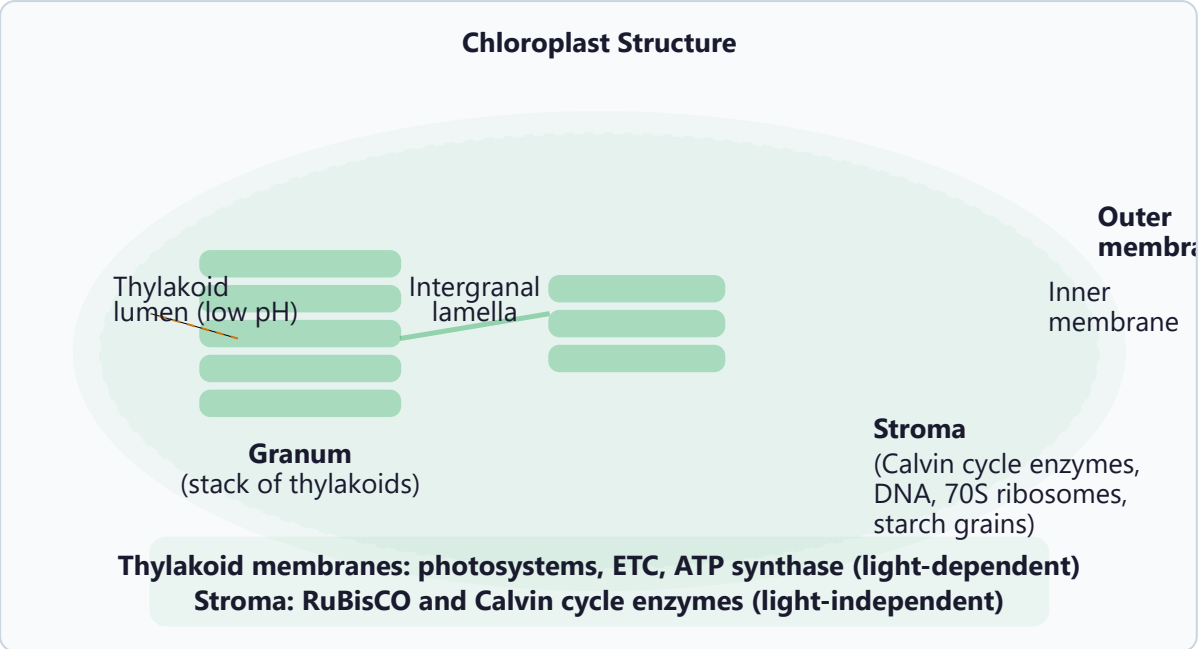
5.1 Photosynthesis

Photosynthesis: The process by which light energy is used to convert carbon dioxide and water into glucose and oxygen. It occurs in **chloroplasts** and involves two main stages: the **light-dependent reactions** (on thylakoid membranes) and the **light-independent reactions** / Calvin cycle (in the stroma). It is the fundamental process that converts light energy into chemical energy (stored in organic molecules), making it the entry point for almost all energy in biological systems.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2 \quad (\Delta G = +2870 \text{ kJ mol}^{-1}, \text{endergonic})$$

Chloroplast Structure

Chloroplasts are double-membrane organelles (~5-10 µm long) with their own circular DNA and 70S ribosomes (evidence for endosymbiotic origin). Their internal structure is highly adapted for photosynthesis:



Photosynthetic Pigments

Chloroplasts contain several pigments organised into **photosystems** (light-harvesting complexes):

Pigment	Colour	Absorption Peaks	Role
Chlorophyll a	Blue-green	430 nm (blue) and 662 nm (red)	Primary pigment; reaction centre of both photosystems; directly involved in light reactions
Chlorophyll b	Yellow-green	453 nm and 642 nm	Accessory pigment; absorbs light and transfers energy to chlorophyll a; broadens absorption range
Carotenoids (carotene, xanthophyll)	Orange/yellow	400-500 nm (blue-violet)	Accessory pigments; also protect chlorophyll from photo-oxidation damage

Pigments can be separated by **chromatography** (see Required Practical 7). The R_f value = distance moved by pigment / distance moved by solvent front.

Exam Tip: Leaves appear green because chlorophyll reflects green light (around 550 nm) and absorbs red and blue light most strongly. The absorption spectrum (wavelengths absorbed) closely

mirrors the action spectrum (wavelengths driving photosynthesis) — this provides evidence that chlorophyll is the main photosynthetic pigment.

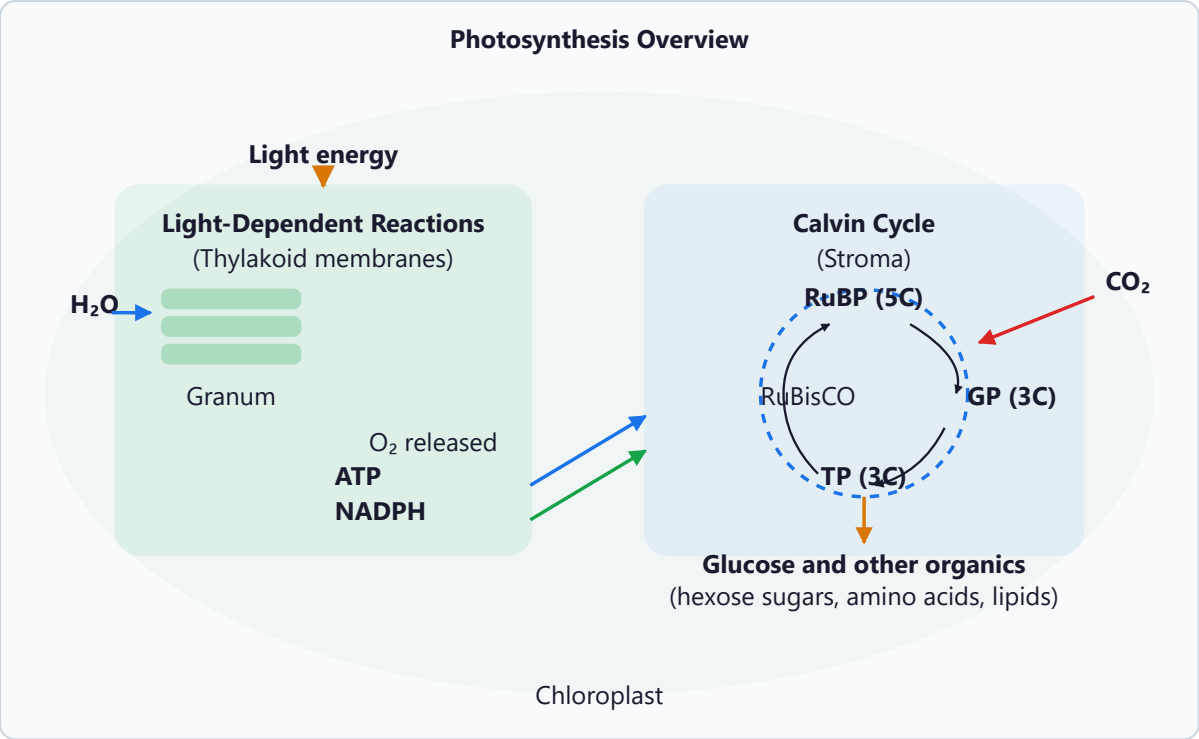
Light-Dependent Reactions (Thylakoid Membranes)

These reactions require light energy directly and produce ATP and reduced NADP (NADPH) for use in the Calvin cycle, plus oxygen as a by-product.

Non-Cyclic Photophosphorylation (the main pathway)

1. **Photoexcitation of PSII:** Light energy is absorbed by chlorophyll and accessory pigments in **photosystem II** (PSII, P680). Energy is funnelled to the reaction centre chlorophyll a molecule. Two electrons in the chlorophyll are excited to a higher energy level and released to an electron acceptor.
2. **Photolysis of water:** The lost electrons are replaced by splitting water (photolysis): $\mathbf{2H_2O \rightarrow 4H^+ + 4e^- + O_2}$. The oxygen is released as a by-product. The H^+ ions remain in the thylakoid lumen, contributing to the proton gradient.
3. **Electron transport chain (ETC):** High-energy electrons pass from PSII along a chain of electron carriers embedded in the thylakoid membrane (plastoquinone, cytochrome b6f complex, plastocyanin). At each carrier, electrons lose energy. This energy is used to **pump H^+ ions from the stroma into the thylakoid lumen**, building a proton (electrochemical) gradient.
4. **Chemiosmosis:** The concentration of H^+ in the lumen is now much higher than in the stroma. H^+ ions flow back through **ATP synthase** (down their concentration gradient). This drives the synthesis of ATP from ADP + P_i. This is called **photophosphorylation** because light energy ultimately drives the phosphorylation.
5. **Photoexcitation of PSI:** The electrons arriving from the ETC enter **photosystem I** (PSI, P700), where light energy re-excites them to an even higher energy level.
6. **Reduction of NADP:** These high-energy electrons are passed to **NADP reductase**, which combines them with H^+ ions (from the stroma) to reduce $NADP^+$ to **NADPH**.

$$\text{Products of non-cyclic photophosphorylation: ATP} + \text{NADPH} + \text{O}_2$$



Cyclic Photophosphorylation

Electrons from PSI are passed back to the electron transport chain between PSII and PSI (rather than to NADP). This produces **ATP only** (no NADPH, no photolysis, no O_2 released). It is used when more ATP than NADPH is needed — the Calvin cycle requires 3 ATP but only 2 NADPH per CO_2 fixed, so cyclic photophosphorylation provides the extra ATP.

Feature	Non-Cyclic	Cyclic
Photosystems used	PSII and PSI	PSI only
Electron pathway	Linear: H_2O → PSII → ETC → PSI → NADP	Circular: PSI → ETC → back to PSI
Products	ATP + NADPH + O_2	ATP only
Photolysis of water	Yes	No

Light-Independent Reactions — The Calvin Cycle (Stroma)

The Calvin cycle uses the ATP and NADPH from the light-dependent reactions to fix CO_2 into organic molecules. It does not directly require light, but it depends on products of the light reactions and therefore stops in the dark.

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1. **Carbon fixation:** CO_2 combines with **ribulose bisphosphate (RuBP, a 5C molecule)** catalysed by the enzyme **RuBisCO** (ribulose bisphosphate carboxylase/oxygenase). This forms an unstable 6C compound that immediately splits into two molecules of **glycerate-3-phosphate (GP, 3C)**.
2. **Reduction:** GP is reduced to **triose phosphate (G3P/TP, 3C)** using ATP (provides energy and phosphate) and NADPH (provides hydrogen/electrons) from the light-dependent reactions. $NADP^+$ and ADP + P_i are recycled back to the light-dependent reactions.
3. **Regeneration of RuBP:** Most TP (5 out of every 6 molecules) is used to regenerate RuBP, using ATP. This allows the cycle to continue.
4. **Product:** One in six TP molecules is the net gain — it exits the cycle and is used to synthesise glucose (two TP molecules combine to form one hexose), amino acids, lipids, and other organic molecules.

$$\text{For every 3 } CO_2 \text{ fixed: 3 RuBP} + 3\ CO_2 \rightarrow 6\ \text{GP} \rightarrow 6\ \text{TP} \rightarrow 1\ \text{TP net gain} + 5\ \text{TP regenerate 3 RuBP}$$

$$\text{Requires: 9 ATP} + 6\ \text{NADPH per 3 } CO_2 \text{ (i.e. 3 ATP} + 2\ \text{NADPH per } CO_2\text{)}$$

Exam Tip: RuBisCO is the most abundant enzyme on Earth (and the most abundant protein). A very common exam question asks what happens to GP, TP, and RuBP levels when conditions suddenly change:

- **Light stops:** No ATP or NADPH produced. GP accumulates (cannot be reduced to TP). TP and RuBP levels fall (TP still used to regenerate RuBP but not replenished; RuBP still used to fix CO_2 but not regenerated).
- **CO_2 removed:** RuBP accumulates (not being used up by fixation). GP falls (not being produced). TP falls (GP not being reduced).
- **CO_2 increased:** GP increases initially (more fixation), then TP increases, then RuBP falls (used up faster than regenerated), until a new equilibrium is reached.

Fate of Triose Phosphate (TP)

TP is the versatile 3-carbon molecule from which the plant synthesises all its organic molecules:

- **Glucose** (2 TP → 1 glucose) — for respiration, cellulose cell walls, starch storage
- **Sucrose** — transport sugar in phloem

- **Starch** — storage polysaccharide in chloroplasts and amyloplasts
- **Amino acids** — TP converted to organic acids, then aminated using nitrate/ammonium ions from roots
- **Fatty acids and glycerol** — for membrane phospholipids and lipid storage
- **Nucleotides** — for DNA and RNA

Limiting Factors in Photosynthesis

At any given time, the rate of photosynthesis is limited by whichever factor is in shortest supply (Blackman's Law of Limiting Factors).

Factor	How It Limits	Practical Notes
Light intensity	Below saturation: rate increases proportionally (more photoexcitation). Above saturation: another factor becomes limiting.	At very high intensities, photoinhibition can damage photosystems. Inverse square law: intensity $\propto 1/d^2$
CO_2 concentration	Substrate for carbon fixation by RuBisCO. At low [CO_2], RuBisCO limits Calvin cycle rate.	Atmospheric CO_2 (~0.04% / 420 ppm) is often the limiting factor in natural conditions. Greenhouses may enrich to 0.1%
Temperature	Affects enzyme activity (RuBisCO, ATP synthase). Too low: slow kinetics. Too high: enzyme denaturation.	Q_{10} ~2 for most enzymes. Optimum typically 25-35°C. Above ~40°C, rate drops sharply.

5.2 Respiration

Respiration: The metabolic process in which organic molecules (mainly glucose) are broken down in a series of enzyme-catalysed reactions to release energy for ATP synthesis. It occurs in all living cells continuously — it is NOT breathing (ventilation).

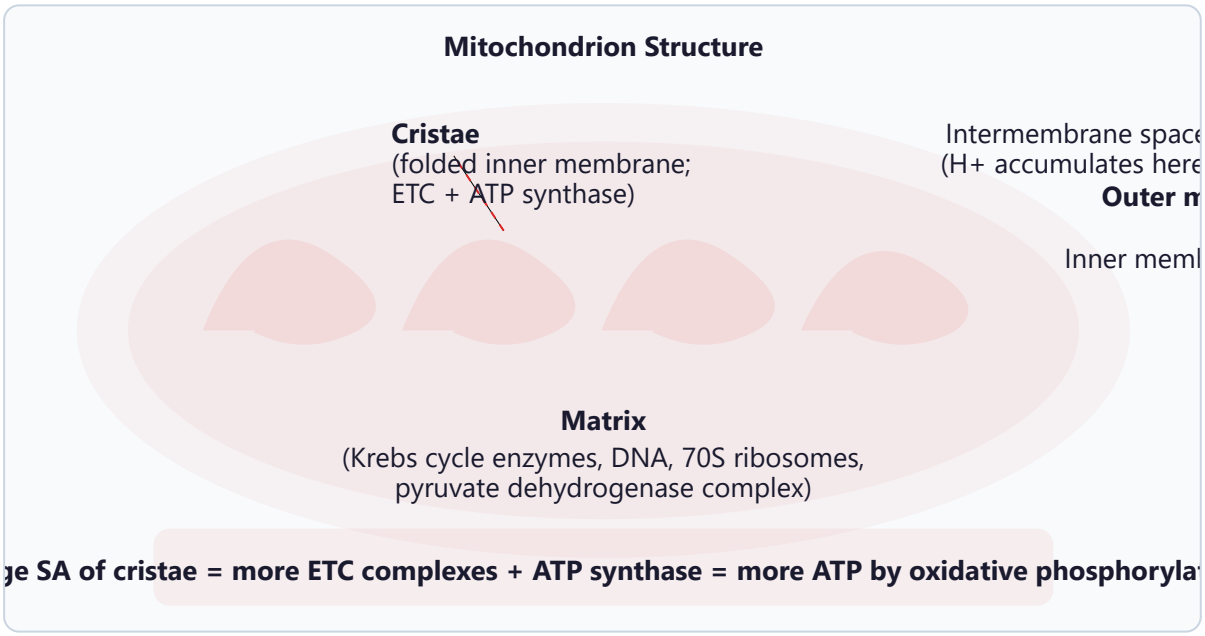
$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \quad (\Delta G = -2870 \text{ kJ mol}^{-1}, \text{exergonic})$$

ATP — The Universal Energy Currency

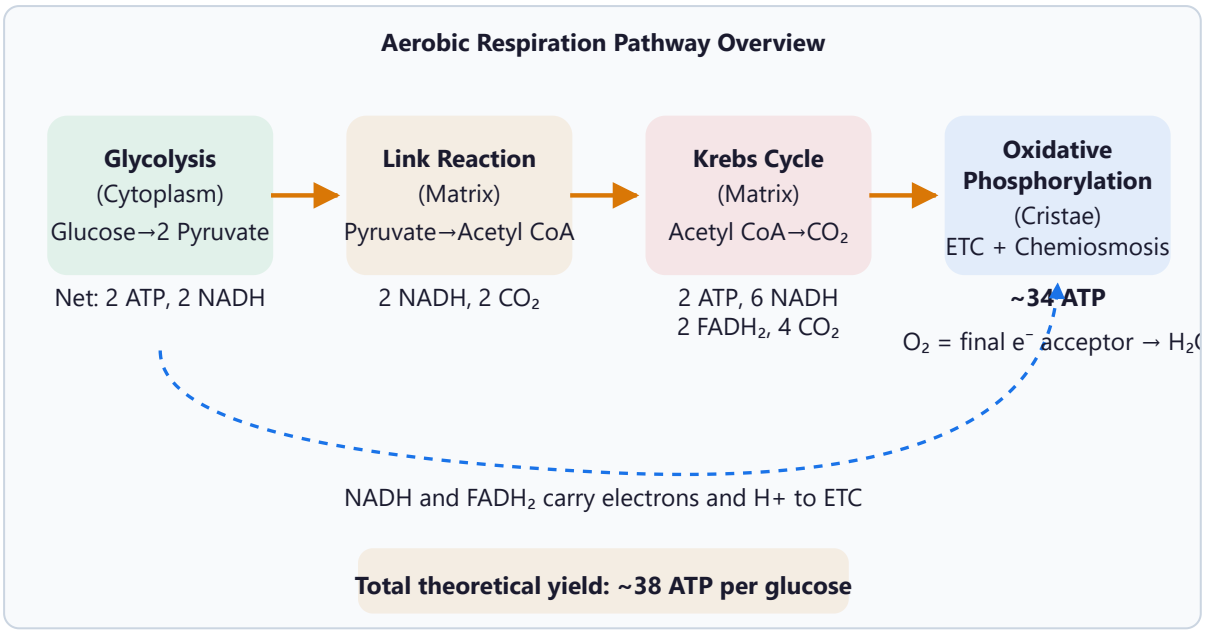
ATP (adenosine triphosphate) is a nucleotide derivative consisting of adenine, ribose, and three phosphate groups. Hydrolysis of ATP to ADP + P_i (catalysed by ATPase) releases ~30.5 kJ mol^{-1} of energy. ATP is ideal as an energy currency because:

- Releases a small, manageable amount of energy per molecule (not wasted as heat)
- Can be rapidly regenerated from ADP + P_i (ATP synthase / substrate-level phosphorylation)
- Soluble and easily transported within cells
- Universal — used by all living organisms
- Cannot pass through cell membranes — each cell makes its own

Mitochondrial Structure



Overview of Aerobic Respiration



Glycolysis (Cytoplasm)

Glycolysis is the splitting of one glucose molecule (6C) into two molecules of pyruvate (3C). It is **anaerobic** (does not require oxygen) and occurs in the cytoplasm of all living cells.

1. **Phosphorylation (energy investment phase):** Glucose is phosphorylated using 2 ATP → fructose-1,6-bisphosphate (6C). This lowers the activation energy for subsequent reactions, makes the molecule more reactive, and prevents it leaving the cell (charged phosphate groups cannot cross membranes).
2. **Lysis:** The 6C fructose-1,6-bisphosphate is split into two molecules of **triose phosphate (3C)** (glyceraldehyde-3-phosphate, G3P).
3. **Oxidation (energy payoff phase):** Each triose phosphate is oxidised to pyruvate (3C). For each triose phosphate: 2 ATP produced by **substrate-level phosphorylation** and 1 NAD^+ reduced to NADH + H^+.

$$\text{Net yield per glucose: } 2\text{ ATP} + 2\text{ NADH} + 2\text{ pyruvate}$$
$$(4\text{ ATP produced - }2\text{ ATP invested} = 2\text{ ATP net gain})$$

Link Reaction (Mitochondrial Matrix)

Pyruvate is actively transported from the cytoplasm into the mitochondrial matrix via a specific carrier protein in the inner membrane. There it undergoes **oxidative decarboxylation**:

- One carbon is removed as CO_2 (decarboxylation)
- NAD^+ is reduced to NADH (oxidation)

- The remaining 2C acetyl group is combined with **coenzyme A (CoA)** to form **acetyl CoA**

$$\text{Pyruvate (3C)} + \text{NAD}^+ + \text{CoA} \rightarrow \text{Acetyl CoA (2C)} + \text{CO}_2 + \text{NADH} \quad \text{(x2 per glucose)}$$

Krebs Cycle (Mitochondrial Matrix)

The Krebs cycle (citric acid cycle) is a series of enzyme-catalysed reactions that completely oxidise the acetyl group from acetyl CoA:

1. **Acetyl CoA (2C)** combines with **oxaloacetate (4C)** to form **citrate (6C)**. CoA is released and recycled.
2. Citrate undergoes a series of **decarboxylation** and **oxidation** reactions, producing two molecules of CO_2 and regenerating oxaloacetate.
3. At various steps, NAD^+ and FAD are reduced to NADH and $FADH_2$ (these carry electrons to the ETC).
4. One molecule of ATP is produced by substrate-level phosphorylation (actually GTP, which is quickly converted to ATP).

$$\text{Per turn of the Krebs cycle: } 1\ \text{ATP} + 3\ \text{NADH} + 1\ \text{FADH}_2 + 2\ \text{CO}_2$$
$$\text{Per glucose (2 turns): } 2\ \text{ATP} + 6\ \text{NADH} + 2\ \text{FADH}_2 + 4\ \text{CO}_2$$

Exam Tip: The main purpose of the Krebs cycle is NOT to produce ATP directly (only 2 per glucose) — it is to generate **reduced coenzymes** (NADH and $FADH_2$) that carry electrons to the electron transport chain, where the vast majority of ATP is produced. The CO_2 you breathe out comes from the link reaction and Krebs cycle (decarboxylation reactions).

Oxidative Phosphorylation (Inner Mitochondrial Membrane — Cristae)

This is where most ATP is produced (~34 out of ~38 total per glucose). It involves the electron transport chain and chemiosmosis:

1. **NADH and $FADH_2$** donate their electrons and H^+ ions to the **electron transport chain (ETC)** — a series of protein complexes embedded in the inner mitochondrial membrane (cristae). NADH donates to Complex I; $FADH_2$ donates to Complex II (at a lower energy level, producing fewer ATP).

2. Electrons pass along the chain of carriers (including ubiquinone, cytochrome c, and cytochrome oxidase), losing energy at each step.
3. The energy released is used to **pump H^+ ions from the matrix into the intermembrane space**, creating a steep electrochemical (proton) gradient.
4. H^+ ions flow back into the matrix through **ATP synthase** (down their concentration gradient). This flow drives the rotation of ATP synthase, catalysing ATP synthesis from ADP + P_i. This process is called **chemiosmosis** (proposed by Peter Mitchell, 1961).
5. **Oxygen is the final electron acceptor.** At the end of the ETC, electrons and H^+ combine with molecular oxygen to form water: $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$. Without oxygen, electrons cannot leave the chain, the ETC stops, NADH cannot be reoxidised, and the Krebs cycle halts.

Exam Tip: Cyanide inhibits cytochrome c oxidase (Complex IV — the final ETC complex). This stops electron flow, H^+ pumping ceases, the proton gradient collapses, and no ATP is made by chemiosmosis. The organism must rely on anaerobic respiration (glycolysis) alone — producing only 2 ATP per glucose, which is insufficient for survival. This is why cyanide is so lethal even in small doses.

Total ATP Yield (Theoretical Maximum per Glucose)

Stage	Location	ATP (net)	NADH	$FADH_2$
Glycolysis	Cytoplasm	2	2	0
Link reaction (x2)	Matrix	0	2	0
Krebs cycle (x2)	Matrix	2	6	2
Oxidative phosphorylation	Cristae	~34 (10 NADH x ~2.5 + 2 $FADH_2$ x ~1.5)	—	—
Total		**~38 ATP**	—	—

Note: The actual yield is less than 38 ATP because: (1) some of the proton gradient energy is used for other transport processes; (2) NADH from glycolysis must be shuttled into the mitochondria, costing

energy; (3) the inner membrane is not perfectly impermeable to H^+. The real yield is approximately **30-32 ATP per glucose**.

Anaerobic Respiration

When oxygen is unavailable, oxidative phosphorylation stops. NAD^+ must be regenerated from NADH for glycolysis to continue (otherwise glycolysis would also halt). Anaerobic pathways regenerate NAD^+ without using oxygen:

Feature	Lactate Fermentation (Animals, some bacteria)	Alcoholic Fermentation (Yeast, some plants)
Reaction	Pyruvate + NADH → Lactate + NAD^+	Pyruvate → Ethanal + CO_2 Ethanal + NADH → Ethanol + NAD^+
Enzyme	Lactate dehydrogenase	Pyruvate decarboxylase, then alcohol dehydrogenase
Reversible?	Yes — lactate can be converted back to pyruvate when O_2 available (in liver)	No — ethanol is toxic; organism cannot convert it back
ATP yield	2 ATP per glucose (glycolysis only)	2 ATP per glucose (glycolysis only)
CO_2 produced?	No	Yes (in decarboxylation step)

Oxygen debt: After vigorous exercise, extra oxygen is consumed to: (1) convert lactate back to pyruvate (and then to glucose via gluconeogenesis in the liver — the Cori cycle); (2) re-oxygenate haemoglobin and myoglobin; (3) replenish phosphocreatine stores in muscle.

Respiratory Substrates

Cells can respire substrates other than glucose:

- **Lipids:** Glycerol enters glycolysis as triose phosphate. Fatty acids undergo beta-oxidation in the matrix to produce acetyl CoA (which enters the Krebs cycle) and reduced NAD/FAD. Lipids yield more ATP per gram than carbohydrates (~9 kcal/g vs ~4 kcal/g) because they are more reduced (more C-H bonds to oxidise).

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- **Proteins:** Amino acids are deaminated (amino group removed as urea). The remaining carbon skeleton enters respiration as pyruvate, acetyl CoA, or a Krebs cycle intermediate, depending on the amino acid.

The **respiratory quotient (RQ)** indicates which substrate is being respired:

$$RQ = CO_2 \text{ produced} / O_2 \text{ consumed}$$

Substrate	RQ
Carbohydrate (glucose)	1.0
Protein	~0.9
Lipid (fat)	~0.7
Anaerobic respiration	> 1.0 (CO_2 produced without O_2 consumption)

5.3 Energy and Ecosystems

Energy Flow Through Ecosystems

Gross Primary Productivity (GPP): The total rate of energy fixation (photosynthesis) by producers in an ecosystem. Measured in kJ m^{-2} $year^{-1}$.

$$\text{Net Primary Productivity (NPP)} = \text{GPP - Respiration (R)}$$

NPP is the rate of energy storage in plant biomass — this is the energy available to consumers

Similarly, for consumers:

$$\text{Net Production (N)} = \text{Ingested energy (I) - Faecal losses (F) - Respiratory losses (R)}$$

Energy transfer between trophic levels is **inefficient** (typically only 5-20% of energy passes to the next level). Energy is lost because:

- **Not all organisms are eaten** — some die and are decomposed (enter the detritivore food chain)
- **Not all parts eaten are digested** — indigestible material (cellulose, bone, chitin) is egested as faeces
- **Respiratory heat loss** — the largest loss at each level; energy released from respiration as metabolic heat cannot be used by the next trophic level (except in endotherms where it maintains body temperature)
- **Excretory losses** — urea in urine, CO_2 breathed out

$$\text{Efficiency of energy transfer} = (\text{Energy available at trophic level n+1} \ / \ \text{Energy available at trophic level n}) \times 100\%$$

Exam Tip: This efficiency is typically 5-20% between trophic levels. This explains why food chains rarely exceed 4-5 trophic levels — too little energy reaches higher levels to sustain populations. It also explains why farming animals is less energy-efficient than growing crops. Endotherms (mammals, birds) lose more energy as heat than ectotherms (reptiles, fish), so conversion efficiency is lower in cattle (~4%) than in fish (~10%).

Farming Practices and Productivity

Agricultural practices aim to maximise NPP (crop yield) or to improve energy transfer efficiency in livestock:

- **Crop productivity:** Fertilisers (replace minerals removed at harvest), pesticides (reduce energy loss to pests), selective breeding for high-yield varieties, glasshouses (optimise limiting factors)
- **Livestock productivity:** Restricting movement (less energy lost in locomotion), controlling temperature (less energy for thermoregulation), high-energy feed (more digestible), selective breeding for fast growth

Nutrient Cycles

The Carbon Cycle:

- CO_2 enters living systems via **photosynthesis** (carbon fixation by RuBisCO in producers)
- Carbon passes through food chains as organic molecules (glucose, amino acids, lipids, nucleic acids)
- CO_2 returned to atmosphere by **respiration** (all organisms), **combustion** of fossil fuels, and **decomposition** (saprobionts release CO_2 through respiration as they break down dead organic matter)

- Carbon stored long-term as fossil fuels (coal, oil, natural gas — formed from ancient organisms), limestone ($CaCO_3$), and dissolved ocean carbonates
- Human combustion of fossil fuels is releasing stored carbon faster than it can be re-fixed, causing rising atmospheric CO_2 and climate change

The Nitrogen Cycle:

- **Nitrogen fixation:** $N_2 \rightarrow NH_4^+$ (ammonium) by *Rhizobium* bacteria in root nodules of leguminous plants (mutualistic relationship). Also by free-living *Azotobacter*, and industrially by the Haber process (for fertiliser). Lightning can also fix small amounts.
- **Nitrification:** $NH_4^+ \rightarrow NO_2^-$ (by *Nitrosomonas*) $\rightarrow NO_3^-$ (by *Nitrobacter*). These are **chemoautotrophic** nitrifying bacteria that require **aerobic conditions**.
- **Assimilation:** Plants absorb NO_3^- or NH_4^+ via active transport in root hairs. These are used to make amino acids, nucleotides, and chlorophyll. Animals obtain nitrogen by eating plants/other animals.
- **Ammonification:** Decomposers (saprobionts) break down organic nitrogen compounds (proteins, nucleic acids) in dead organisms and excretory products $\rightarrow NH_4^+$ released into soil.
- **Denitrification:** $NO_3^- \rightarrow N_2$ (gas lost to atmosphere) by denitrifying bacteria (e.g. *Pseudomonas*) under **anaerobic conditions** (waterlogged soil). This is a loss of useful nitrogen from the soil.

Exam Tip: Remember the conditions: nitrification requires aerobic conditions (well-aerated soil), while denitrification occurs under anaerobic conditions (waterlogged soil). Farmers plough soil to increase aeration, promoting nitrification and reducing denitrification — this keeps nitrogen available as nitrate for plant uptake.

The Phosphorus Cycle:

- Phosphorus is found in rocks as phosphate minerals; released by **weathering** over geological timescales
- Phosphate ions (PO_4^{3-}) absorbed by plants from soil solution via active transport
- Passes through food chains in nucleic acids, ATP, phospholipids, bone
- Returned to soil by decomposition of dead organisms and excretion
- Leached into rivers and eventually deposited in ocean sediments; re-incorporated into sedimentary rock over millions of years

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- Phosphorus has no gaseous phase — unlike carbon and nitrogen, it does not cycle through the atmosphere

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